Day 050 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Use Factors and Multiples to Explain Structure

Standards: 6.NS.B.4a, 6.NS.B.4b

Learning target: I can use Factors and Multiples to Explain Structure, and explain and check my reasoning.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. List factor pairs or use prime factorizations. The greatest factor shared by both numbers is the GCF. For identical kits with no leftovers, the GCF gives the greatest number of kits. List positive multiples of both numbers. The smallest shared multiple is the LCM. Repeating events that begin together next occur together after this many time units.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 17

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01 / Learn

List factor pairs or use prime factorizations. The greatest factor shared by both numbers is the GCF. For identical kits with no leftovers, the GCF gives the greatest number of kits. List positive multiples of both numbers. The smallest shared multiple is the LCM. Repeating events that begin together next occur together after this many time units.

Worked example 1

Find the GCF of 8 and 12. Show factor pairs or prime factors.

1. List factor pairs or use prime factorizations.

2. The greatest factor shared by both numbers is the GCF.

3. Common factors: 1, 2, 4

4. greatest is 4.

5. State the result in the form requested: 4.

Answer: 4

Worked example 2

Two signals flash together now and then every 2 and 9 seconds. Select all true statements.
A. They next flash together after 18 seconds.
B. They next flash together after 1 seconds.
C. 36 seconds is also a time they flash together.
D. The next shared time is 19 seconds.

1. List positive multiples of both numbers.

2. List multiples or use prime factors.

3. The least shared positive multiple is 18

4. multiples of 18 are later shared times.

Answer: A, C - They next flash together after 18 seconds.; 36 seconds is also a time they flash together.

Day 050 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find the GCF of 12 and 16. Show factor pairs or prime factors.

02 / Constructed response

Find the LCM of 5 and 4 by listing multiples.

03 / Constructed response

Find the GCF of 6 and 9. Show factor pairs or prime factors.

04 / Constructed response

Find the LCM of 12 and 6 by listing multiples.

05 / Constructed response

Find the GCF of 60 and 84. Show factor pairs or prime factors.

06 / Constructed response

Find the LCM of 3 and 8 by listing multiples.

07 / Constructed response

Make the greatest possible number of identical kits from 70 pencils and 100 erasers with no leftovers. Find the number of kits and each kit’s contents.

08 / Multiple select

Two signals flash together now and then every 3 and 11 seconds. Select all true statements.
A. 66 seconds is also a time they flash together.
B. They next flash together after 1 seconds.
C. The next shared time is 34 seconds.
D. They next flash together after 33 seconds.

09 / Multiple select

Select all true statements about 36 and 54. Explain why the greatest factor fits the kit situation.
A. 90 divides both counts.
B. 19 is their greatest common factor.
C. 18 divides both 36 and 54.
D. 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

10 / Constructed response

Two lights flash together now, then every 8 and 3 seconds. When do they next flash together? Explain.

Day 050 | Unit 04 | Answer key

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1. 4

Common factors: 1, 2, 4; greatest is 4.

6.NS.B.4a | Constructed response

2. 20

The first positive multiple shared by 5 and 4 is 20.

6.NS.B.4b | Constructed response

3. 3

Common factors: 1, 3; greatest is 3.

6.NS.B.4a | Constructed response

4. 12

The first positive multiple shared by 12 and 6 is 12.

6.NS.B.4b | Constructed response

5. 12

Common factors: 1, 2, 3, 4, 6, 12; greatest is 12.

6.NS.B.4a | Constructed response

6. 24

The first positive multiple shared by 3 and 8 is 24.

6.NS.B.4b | Constructed response

7. 10 kits; 7 pencils and 10 erasers each

GCF(70, 100) = 10; divide both counts by 10.

6.NS.B.4a | Constructed response

8. A, D - 66 seconds is also a time they flash together.; They next flash together after 33 seconds.

List multiples or use prime factors. The least shared positive multiple is 33; multiples of 33 are later shared times.

6.NS.B.4b | Multiple select

9. C, D - 18 divides both 36 and 54.; 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

GCF(36, 54) = 18. Divide each count by 18: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

10. 24 seconds

The first positive multiple shared by 8 and 3 is 24.

6.NS.B.4b | Constructed response